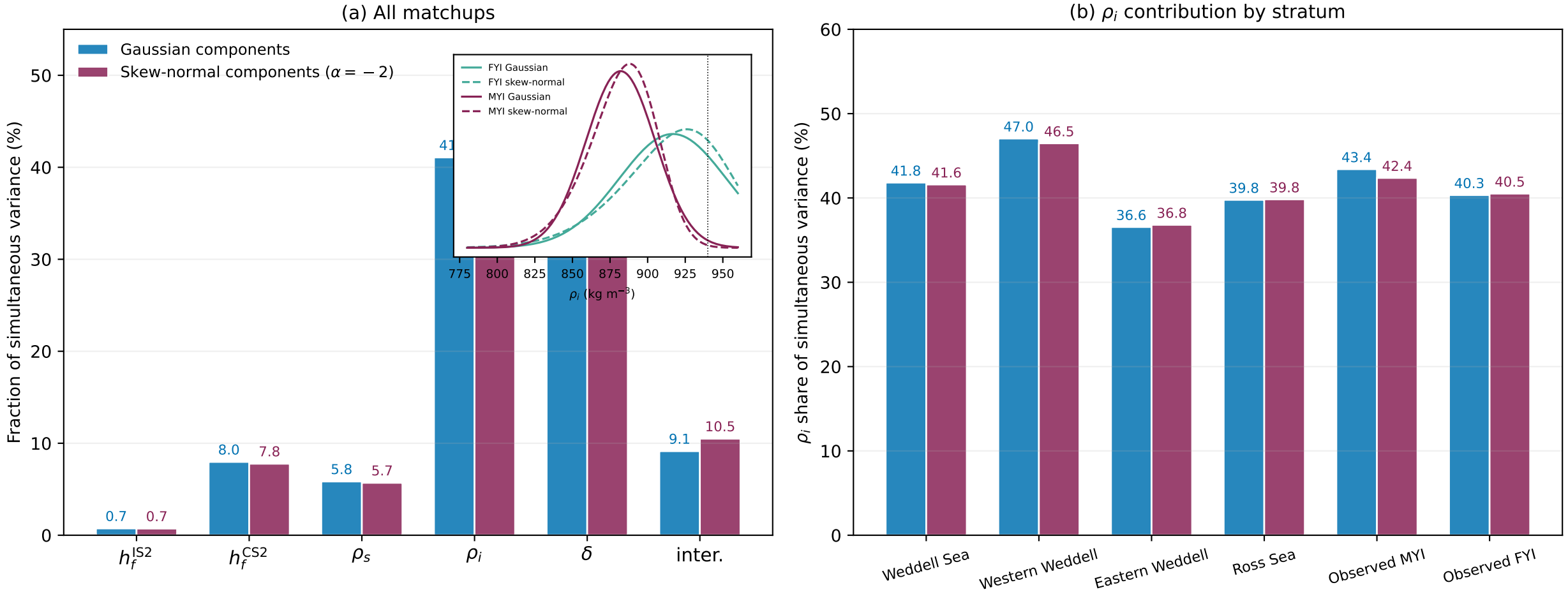
Robustness of the variance decomposition to the shape of the ρ_i mixture components (940 kg m^{-3} cap and MYI weight unchanged)
(a) All matchups
Gaussian components
Skew-normal components ($\alpha = -2$)
Fraction of simultaneous variance (%)
0
10
20
30
40
50
0.7
0.7
8.0
7.8
5.8
5.7
41
9.1
10.5
h_f^{IS2}
h_f^{CS2}
ρ_s
ρ_i
δ
inter.
FYI Gaussian
FYI skew-normal
MYI Gaussian
MYI skew-normal
775
800
825
850
875
900
925
950
ρ_i (kg m^{-3})
(b) ρ_i contribution by stratum
ρ_i share of simultaneous variance (%)
0
10
20
30
40
50
60
41.8
41.6
47.0
46.5
36.6
36.8
39.8
39.8
43.4
42.4
40.3
40.5
Weddell Sea
Western Weddell
Eastern Weddell
Ross Sea
Observed MYI
Observed FYI